An Enhanced Adaptive RTT-Fair Congestion Control Mechanism for BBR Using Smoothed RTT and Nonlinear BDP Adaptation

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Abstract

The Bottleneck Bandwidth and Round-trip propagation time (BBR) congestion control algorithm is designed to achieve high network utilization while maintaining low queuing delay. However, when multiple BBR flows with different Round-Trip Times (RTTs) share a common bottleneck, significant RTT unfairness can occur, allowing flows with larger RTTs and higher Bandwidth-Delay Products (BDPs) to obtain a disproportionate share of the available bandwidth. The FaiRTT algorithm addresses this issue by dynamically adjusting the estimated BDP and in-flight data based on RTT measurements, thereby improving fairness between competing flows. This project proposes an enhanced version of FaiRTT, referred to as Enhanced FaiRTT, with the objective of improving RTT fairness while preserving bottleneck throughput and network utilization. The proposed approach introduces three major improvements. First, instantaneous RTTmeasurements are replaced with a smoothed RTT estimation to reduce the effect of shortterm RTT fluctuations and provide more stable congestion-control decisions. Second, the fairness and discount parameters used by FaiRTT are made adaptive rather than fixed, allowing the algorithm to respond dynamically to changes in congestion and RTT differences among competing flows. Third, a nonlinear and

bounded BDP adjustment mechanism is introduced to prevent overly aggressive reduction of in-flight data when large RTT differences occur, thereby maintaining a balance between fairness and throughput. The proposed algorithm will be implemented and evaluated using the NS-3 network simulation framework. Its performance will be compared with BBRv2, the original FaiRTT algorithm, and BBRv3 under different network conditions, including varying RTTs, bottleneck queue sizes, and numbers of competing flows. The evaluation will consider Jain's fairness index, per-flow and aggregate throughput, bottleneck link utilization, RTT variation, packet loss, retransmissions, queue occupancy, and convergence behavior. The proposed work aims to provide a more stable and adaptive RTT-fair congestion-control mechanism for BBR while maintaining efficient utilization of the available bottleneck bandwidth. Keywords: BBR,FaiRTT,RTTFairness,CongestionControl, Bandwidth-DelayProduct, Smoothed RTT, Adaptive Congestion Control, NS-3.

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